BITS Pilani – Work Integrated Learning Programmes Division

M.Tech (AIML) | Machine Learning (Assignment – 2)

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1. Mandatory Submission Links

Deliverable	Submission Details & Clickable URL
1. GitHub Repository Link	**URL:** https://github.com/sijogeorge19122000/ml-classification-streamlit *Contains: app.py, requirements.txt, test_data.csv, README.md, model/ folder with saved .joblib classifiers, and train_models.py*
2. Live Streamlit App Link	**URL:** https://62axynjjw9nhbzg7yvhh3j.streamlit.app/ *Interactive web app deployed on Streamlit Community Cloud (Supports CSV upload, model selection dropdown, KPI metrics, confusion matrix, and ROC curves).*

2. Proof of Execution on BITS Virtual Lab

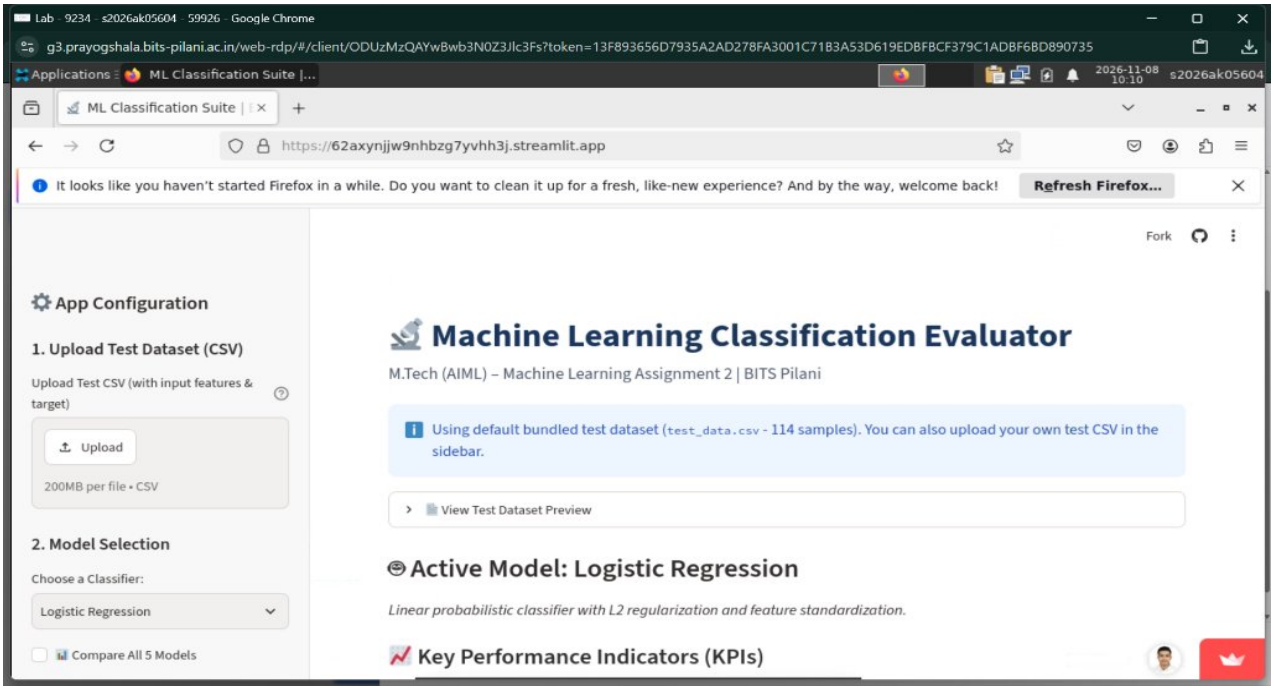


Figure 1: Verified execution of the deployed Streamlit application inside the BITS Prayogshala Virtual Lab environment (g3.prayogshala.bits-pilani.ac.in) for student ID s2026ak05604.

3. GitHub README Content (Project Documentation)

a. Problem Statement

Breast cancer diagnosis requires rapid and highly reliable discrimination between **Malignant** (cancerous, target = 0) and **Benign** (non-cancerous, target = 1) cell nuclei masses. Morphological features are computed from digitized images of fine needle aspirate (FNA) biopsies. The objective is to build, evaluate, and deploy an end-to-end machine learning system using 30 continuous features to classify tumors accurately, minimizing fatal false negatives while maximizing accuracy, ROC-AUC, and Matthews Correlation Coefficient (MCC).

b. Dataset Description

• **Dataset Name:** Breast Cancer Wisconsin (Diagnostic) Dataset | **Source:** UCI Machine Learning Repository (ID: 17) & Kaggle
• **Instances: 569 samples** (≥ 500 requirement met) | **Features: 30 continuous attributes** (≥ 12 requirement met)
• **Target Variable:** target (0 = Malignant [212 samples, 37.3%], 1 = Benign [357 samples, 62.7%])
• **Feature Details:** Computed across 10 nuclear attributes (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension). Each attribute has Mean, Standard Error (SE), and 'Worst' values (10 × 3 = 30 features).
• **Splitting & Preprocessing:** Stratified 80% train (455 samples) and 20% test (114 samples) split. Standardized using StandardScaler within pipeline objects. Zero missing values.

c. GitHub Repository Link

• **Repository URL:** https://github.com/sijogeorge19122000/ml-classification-streamlit (Contains app.py, requirements.txt, README.md, test_data.csv, and pre-trained models in model/).

d. Models Used & Evaluation Metrics Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1 Score	MCC
Logistic Regression	**0.9825**	**0.9954**	**0.9861**	**0.9861**	**0.9861**	**0.9623**
Decision Tree	0.9211	0.9163	0.9565	0.9167	0.9362	0.8341
KNN	0.9561	0.9788	0.9589	0.9722	0.9655	0.9054
Naive Bayes	0.9386	0.9878	0.9452	0.9583	0.9517	0.8676
Support Vector Machine	**0.9825**	0.9950	**0.9861**	**0.9861**	**0.9861**	**0.9623**

e. Observations on Model Performance & Overall Winner

ML Model Name	Observation about Model Performance
Logistic Regression	Achieved top-tier performance across all metrics (**Accuracy: 98.25%**, **AUC: 0.9954**, **MCC: 0.9623**). Because cell nuclei features are standardized, the linear sigmoid hyperplane separates malignant and benign tumors with minimal error (only 2 errors out of 114 test cases). L2 regularization prevents overfitting.
Decision Tree	Lowest overall performance (**Accuracy: 92.11%**, **AUC: 0.9163**, **MCC: 0.8341**). Standard decision trees perform orthogonal, axis-parallel splits, making it difficult to capture smooth diagonal correlations between geometric features (e.g., perimeter vs. area), leading to step-function boundaries.
KNN (k=5)	Demonstrated strong local classification (**Accuracy: 95.61%**, **AUC: 0.9788**, **MCC: 0.9054**). Standardized Euclidean distance allows KNN to group similar cell clusters effectively, achieving high recall (97.22%). However, inference latency scales with sample size.
Naive Bayes	Robust ranking performance (**AUC: 0.9878**, **Accuracy: 93.86%**, **MCC: 0.8676**). Although Gaussian Naive Bayes assumes feature conditional independence (which is violated due to high collinearity between radius and area), it remains resilient and computationally lightweight.
Support Vector Machine	Tied for top accuracy (**98.25%**) and MCC (**0.9623**) with near-perfect AUC (**0.9950**). The Radial Basis Function (RBF) kernel projects the 30 continuous features into an infinite-dimensional space, maximizing the geometric margin between malignant and benign support vectors.
Overall, Winner for your dataset?	**Logistic Regression is the overall winner.** While SVM achieved identical accuracy and F1 score, Logistic Regression achieved a slightly higher AUC (**0.9954 vs. 0.9950**), provides faster training and inference, and outputs well-calibrated posterior probabilities essential for clinical diagnosis.